



Building Agentic AI for Full HR Lifecycle

12 Governed HR Agents | RAG | Tool Calling | Workflow Orchestration
Human-in-the-Loop | Evaluation | Responsible AI

Portfolio objective

A practical agent-design portfolio showing how Agentic AI can support the full HR lifecycle - from employee services and recruitment through assessment, talent, performance, workforce planning, reward, organisational restructuring, People Analytics and HR Operations.

Common agent architecture

Trigger -> Goal -> Approved Inputs/Data -> Knowledge/RAG -> Reasoning/Orchestration -> Tools/Actions -> Human Approval -> Exception Handling -> Output/Action -> Audit Trail -> Evaluation

What makes this different from a chatbot

Each design combines an explicit goal, governed knowledge, stateful workflow logic, permitted tools/actions, human checkpoints, exception handling, auditability and measurable evaluation. The agent can coordinate multiple steps but cannot bypass HR decision rights.

Technology design environment

Enterprise LLMs / Microsoft Copilot; RAG; structured JSON; tool/function calling; Copilot Studio; Power Automate; Python where appropriate; Power BI / validated analytics; existing HR platforms as governed data/workflow environments.

Responsible AI boundary

No autonomous hiring/rejection, performance rating, HiPo designation, promotion, succession, redundancy/termination, individual compensation or psychological/medical suitability decisions. Technical HRIS implementation/configuration, migration and integration remain outside the claimed scope.

01 HR Policy & Employee Services Agent

Agent goal

Resolve routine HR policy/service requests using approved knowledge, initiate permitted workflows, and escalate exceptions.

Approved inputs / knowledge

Employee query; Approved HR policy/knowledge; Employee context permitted for the request.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Knowledge retrieval; Case/request creation; Workflow routing; Status lookup. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

HR validates ambiguous policy interpretation, sensitive cases and exceptions.

Evaluation

Grounded-answer accuracy; Correct escalation; Unsupported-claim rate; Case-routing accuracy.

02 Recruitment Intelligence & Sourcing Agent

Agent goal

Convert an approved hiring brief into structured search intelligence, talent-pool hypotheses and evidence for recruiter review.

Approved inputs / knowledge

Approved job brief; Role/skills taxonomy; Permitted talent-market evidence.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Brief structuring; Skills normalization; Market-map preparation; Evidence summarization. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Recruiter approves criteria, outreach targets, shortlist decisions and candidate communications.

Evaluation

Brief extraction accuracy; Evidence completeness; Research correction rate; Recruiter acceptance.

03 Job Drafting & Job Architecture Agent

Agent goal

Draft and quality-check job descriptions against approved job architecture, capability language and inclusive-writing rules.

Approved inputs / knowledge

Approved role requirements; Job-family framework; Grade/level guidance; Approved templates.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

JD drafting; Skills extraction; Template validation; Language quality checks. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

HR/job owner approves job scope, level, accountabilities and final published description.

Evaluation

Template compliance; Human edit rate; Missing-requirement rate; Architecture alignment.

04 Onboarding Coordination Agent

Agent goal

Coordinate approved onboarding actions, monitor completion, answer grounded questions and escalate delays.

Approved inputs / knowledge

New-joiner workflow; Approved onboarding content; Task status.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Task creation; Reminder workflow; Knowledge retrieval; Status summarization. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

HR/manager resolves exceptions and approves employment-related changes.

Evaluation

On-time task completion; Escalation precision; FAQ grounding; Overdue-action reduction.

05 Talent Assessment Agent

Agent goal

Structure assessment evidence, map observations to approved competencies and prepare assessor summaries without making assessment decisions.

Approved inputs / knowledge

Approved competency framework; Assessor observations; Assessment exercise evidence.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Evidence classification; Competency mapping; Report drafting; Evidence-gap flagging. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Qualified assessors validate evidence, assign ratings and make assessment conclusions.

Evaluation

Evidence-to-competency accuracy; Unsupported inference rate; Assessor correction rate; Report completeness.

06 Talent Management & Succession Agent

Agent goal

Prepare talent-review and succession evidence, development actions and mobility options for human calibration.

Approved inputs / knowledge

Validated talent data; Success profiles; Development history; Approved mobility opportunities.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Evidence summarization; Readiness evidence organization; Development option retrieval; Action tracking. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Leaders/HR decide potential, succession, promotion, readiness and mobility outcomes.

Evaluation

Evidence completeness; Calibration usefulness; Action follow-through; AI correction/override rate.

07 Performance Management Agent

Agent goal

Support goal quality, cycle coordination, feedback synthesis and post-rating development planning.

Approved inputs / knowledge

Approved goals framework; Validated feedback; Finalized performance outcomes where applicable.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Goal quality checks; Reminder workflow; Feedback summarization; Development drafting. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Managers set goals and ratings; AI never assigns or recommends performance ratings.

Evaluation

Goal-quality improvement; Summary correction rate; Cycle completion; Development-action uptake.

08 Strategic Workforce Planning Agent

Agent goal

Prepare demand/supply scenarios, skills gaps and Build-Buy-Borrow-Move-Automate options from validated planning assumptions.

Approved inputs / knowledge

Business demand assumptions; Workforce baseline; Skills data; External talent intelligence.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Scenario calculation support; Skills normalization; Gap analysis; Executive commentary drafting. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

HR/business leaders approve assumptions, workforce scenarios and workforce actions.

Evaluation

Scenario traceability; Assumption completeness; Calculation validation; Decision-usefulness rating.

09 Compensation & Benefits Intelligence Agent

Agent goal

Support reward benchmarking, policy retrieval, aggregate pay analysis and scenario preparation without making individual pay decisions.

Approved inputs / knowledge

Approved C&B; policies; Validated aggregate pay data; Licensed benchmark evidence where available.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Policy retrieval; Benchmark evidence structuring; Scenario preparation; Aggregate anomaly flagging. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Reward/HR leaders approve compensation, benefits and individual employee decisions.

Evaluation

Source traceability; Scenario calculation validation; Policy-grounding accuracy; Human correction rate.

10 Organisational Restructuring & Workforce Scenario Agent

Agent goal

Model alternative structures, spans/layers, role families, workforce cost and capability scenarios for leadership review.

Approved inputs / knowledge

Current approved organisation data; Target operating assumptions; Role/capability data; Cost assumptions.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Scenario modelling; Span/layer analysis; Role-overlap analysis; Decision-pack drafting. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

Leadership/HR/legal make restructuring, role-impact, redundancy and employment decisions.

Evaluation

Scenario traceability; Cost-model validation; Role-map completeness; Unsupported recommendation rate.

11 People Analytics & Executive Insight Agent

Agent goal

Interpret validated aggregate HR metrics, investigate defined patterns and prepare executive workforce briefings.

Approved inputs / knowledge

Validated KPI layer; Approved aggregate datasets; Metric definitions.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Metric retrieval; Pattern summarization; Narrative drafting; Threshold alerting. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

People Analytics/HR validates interpretation before executive distribution.

Evaluation

Metric fidelity; Narrative correction rate; Unsupported insight rate; Briefing usefulness.

12 HR Operations Automation Agent

Agent goal

Classify HR requests, retrieve policy, execute permitted workflow steps and escalate exceptions with an audit trail.

Approved inputs / knowledge

Service request; Approved service catalogue; Policy knowledge; Workflow state.

Reasoning & orchestration

Validate trigger and inputs -> retrieve approved evidence -> classify task/risk -> apply deterministic rules or AI reasoning as appropriate -> use permitted tools -> request human approval where required -> execute or escalate -> log and evaluate.

Tools / actions

Intent classification; Knowledge retrieval; Workflow execution; Case update; Notification. Technology can include governed LLM/RAG, Copilot Studio, Power Automate, Python or Power BI where relevant.

Human checkpoint

HR approves sensitive/high-impact cases and handles exceptions.

Evaluation

Classification accuracy; Workflow success; Escalation accuracy; SLA improvement; Audit completeness.